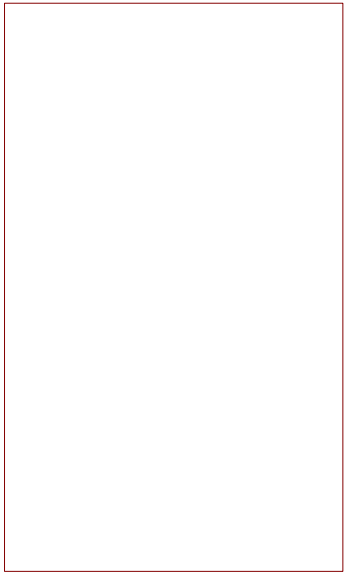
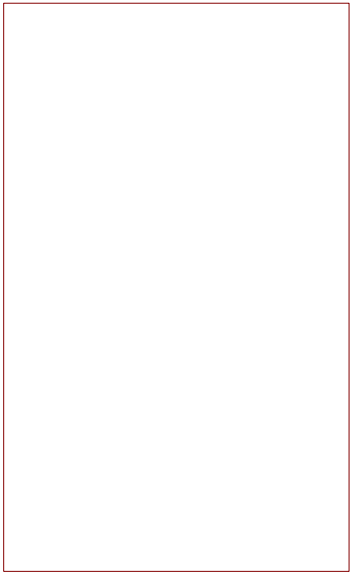
Panel1_MCU



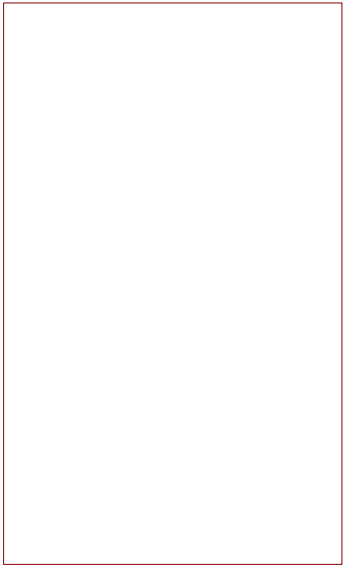
Fájl: Panel1_MCU.kicad_sch

Panel2_COM



Fájl: Panel2_COM.kicad_sch

Panel3_GLM-PSU



Fájl: Panel3_GLM-PSU.kicad_sch

Panel5_FLIR



Fájl: Panel5_FLIR.kicad_sch

Panel4_BAT_Plate



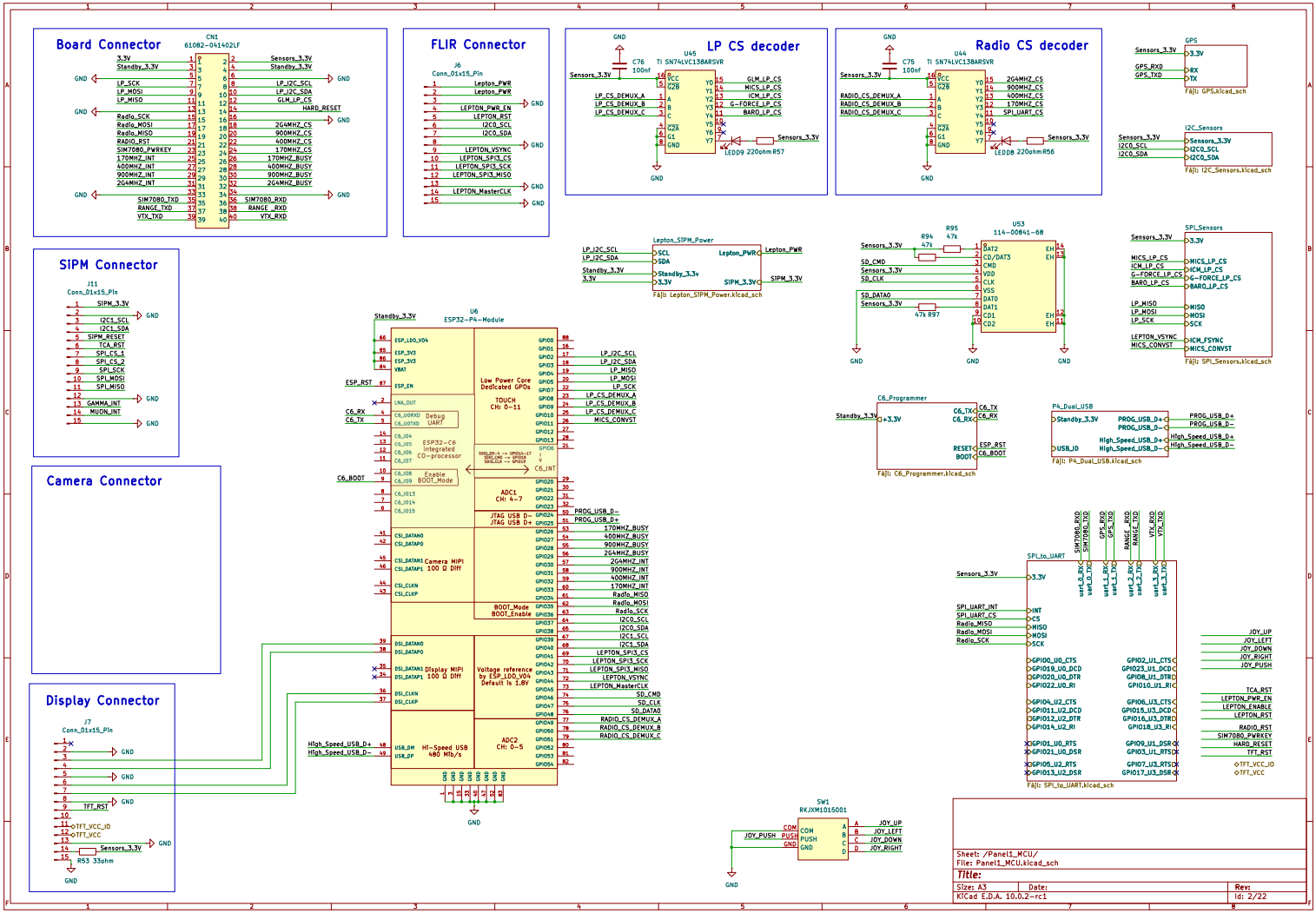
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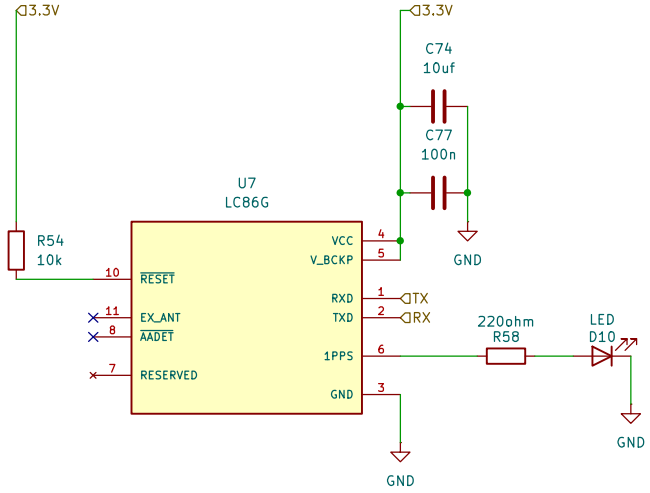
python -m easyeda2kicad --full --lcsc_id C19268026 --output "D:\Data\T.E.E.M.O\Circuit\CanSat_V2\panel

Sheet: /
File: CanSat.kicad_sch

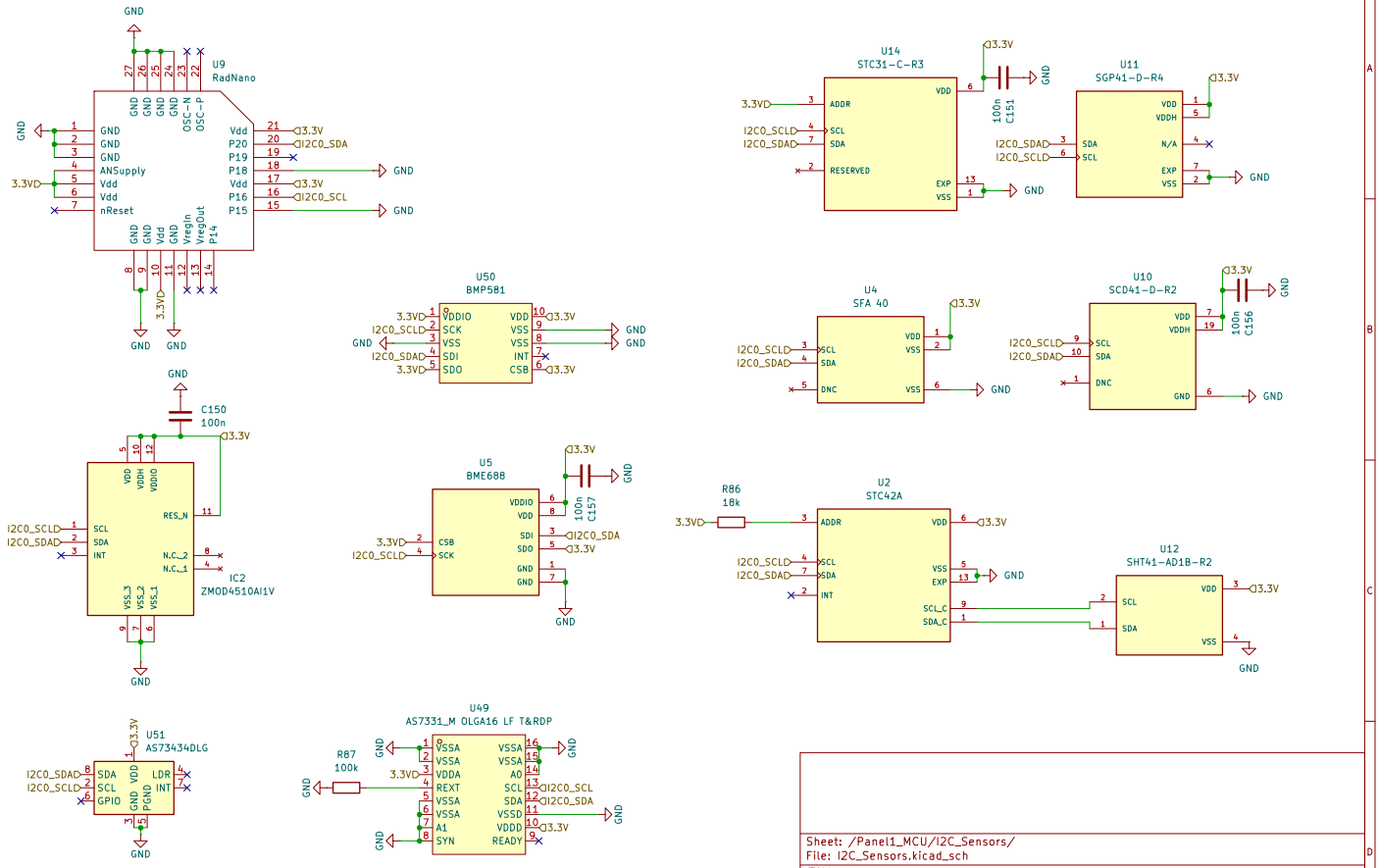
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 1/22

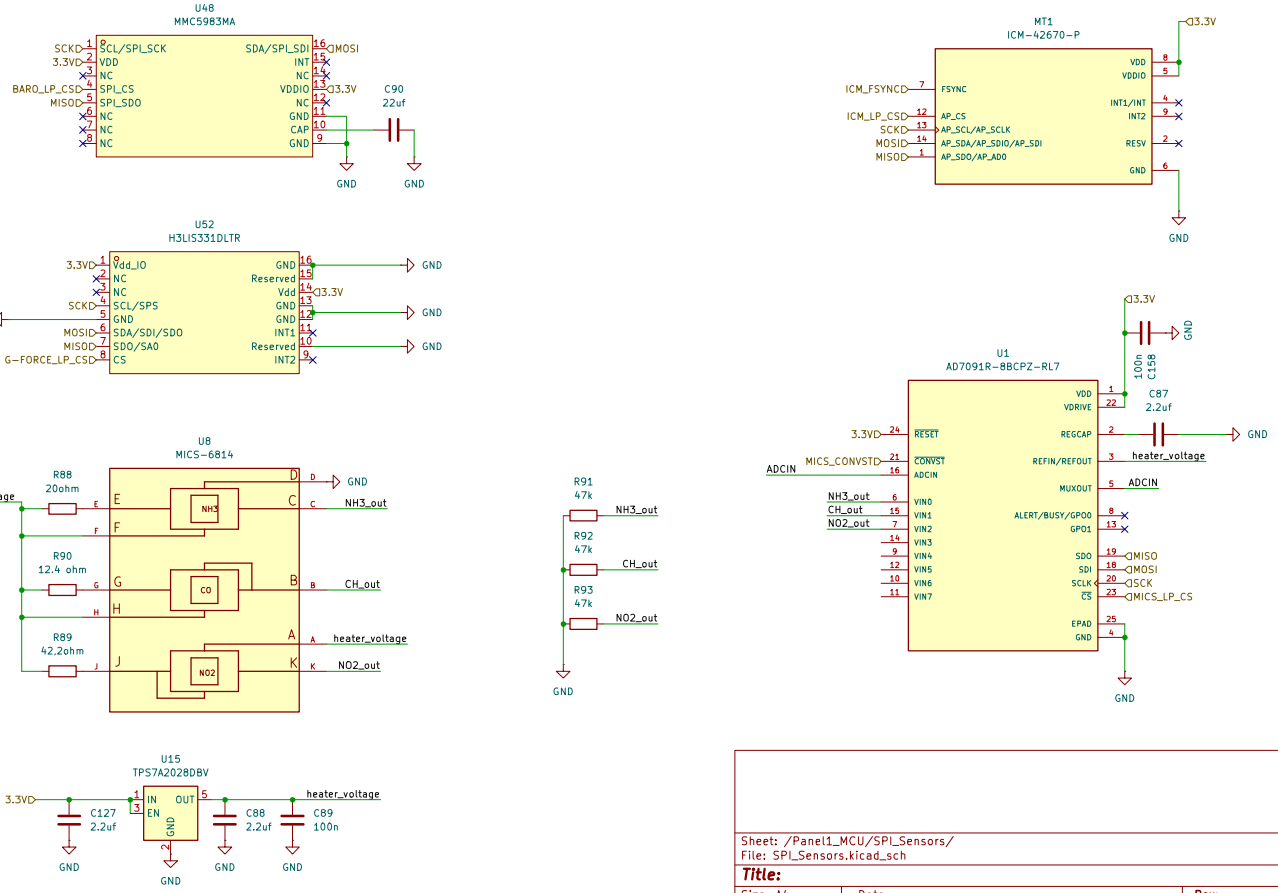

Board Connector
CN1
61082-041402LF
3.3V
Standby_3.3V
GND
LP_SCK
LP_MOSI
LP_MISO
Radio_SCK
Radio_MOSI
Radio_MISO
RADIO_RST
SIM7080_PWRKEY
170MHZ_INT
400MHZ_INT
900MHZ_INT
2G4MHZ_INT
SIM7080_TXD
RANGE_TXD
VTX_TXD
Sensors_3.3V
Standby_3.3V
LP_I2C_SCL
LP_I2C_SDA
GLM_LP_CS
HARD_RESET
2G4MHZ_CS
900MHZ_CS
400MHZ_CS
170MHZ_CS
170MHZ_BUSY
400MHZ_BUSY
900MHZ_BUSY
2G4MHZ_BUSY
SIM7080_RXD
RANGE_RXD
VTX_RXD
FLIR Connector
J6
Conn_01x15_Pin
Lepton_PWR
Lepton_PWR
LEPTON_PWR_EN
LEPTON_RST
I2C0_SCL
I2C0_SDA
LEPTON_VSYNC
LEPTON_SPI3_CS
LEPTON_SPI3_MISO
LEPTON_MasterCLK
LP CS decoder
U45
TI SN74LVC138ARSVR
C76
100nf
Sensors_3.3V
LP_CS_DEMUX_A
LP_CS_DEMUX_B
LP_CS_DEMUX_C
GLM_LP_CS
MICS_LP_CS
ICM_LP_CS
G-FORCE_LP_CS
BARO_LP_CS
LEDD9 220ohm R57
LEDD8 220ohm R56
Radio CS decoder
U44
TI SN74LVC138ARSVR
C75
100nf
RADIO_CS_DEMUX_A
RADIO_CS_DEMUX_B
RADIO_CS_DEMUX_C
2G4MHZ_CS
900MHZ_CS
400MHZ_CS
170MHZ_CS
SPI_UART_CS
GPS
GPS_RXD
GPS_TXD
Fajl: GPS.kicad_sch
I2C_Sensors
I2C0_SCL
I2C0_SDA
Fajl: I2C_Sensors.kicad_sch
SIPM Connector
J11
Conn_01x15_Pin
SIPM_3.3V
I2C1_SCL
I2C1_SDA
SIPM_RESET
TCA_RST
SPI_CS_1
SPI_CS_2
SPI_SCK
SPI_MOSI
SPI_MISO
GAMMA_INT
MUON_INT
Lepton_SIPM_Power
LP_I2C_SCL
LP_I2C_SDA
Standby_3.3V
3.3V
SCL
SDA
Lepton_PWR
SIPM_3.3V
Fajl: Lepton_SIPM_Power.kicad_sch
U53
114-00841-68
R94
47k
R95
47k
R97
47k
SD_CMD
SD_CLK
SD_DATA0
SPI_Sensors
MICS_LP_CS
ICM_LP_CS
G-FORCE_LP_CS
BARO_LP_CS
LP_MISO
LP_MOSI
LP_SCK
LEPTON_VSYNC
MICS_CONVST
ICM_FSYNC
Fajl: SPI_Sensors.kicad_sch
Camera Connector
U6
ESP32-P4-Module
Standby_3.3V
ESP_RST
C6_RX
C6_TX
C6_BOOT
Low Power Core
Dedicated GPOs
TOUCH
CH: 0~11
Debug
UART
ESP32-C6
Integrated
CO-processor
Enable
BOOT_Mode
ADC1
CH: 4~7
JTAG USB D-
JTAG USB D+
Camera MIPI
100 Ω Diff
BOOT_Mode
BOOT_Enable
Display MIPI
100 Ω Diff
Voltage reference
by ESP_LDO_VO4
Default is 1.8V
ADC2
CH: 0~5
USB_DM
USB_DP
HI-Speed USB
480 Mb/s
High_Speed_USB_D+
High_Speed_USB_D-
C6_Programmer
C6_TX
C6_RX
RESET
BOOT
Fajl: C6_Programmer.kicad_sch
P4_Dual_USB
PROG_USB_D+
PROG_USB_D-
High_Speed_USB_D+
High_Speed_USB_D-
USB_ID
Fajl: P4_Dual_USB.kicad_sch
SPI_to_UART
SIM7080_RXD
SIM7080_TXD
GPS_RXD
GPS_TXD
RANGE_RXD
RANGE_TXD
VTX_RXD
VTX_TXD
SPI_UART_INT
SPI_UART_CS
Radio_MISO
Radio_MOSI
Radio_SCK
JOY_UP
JOY_LEFT
JOY_DOWN
JOY_RIGHT
JOY_PUSH
TCA_RST
LEPTON_PWR_EN
LEPTON_ENABLE
LEPTON_RST
RADIO_RST
SIM7080_PWRKEY
HARD_RESET
TFT_RST
TFT_VCC_IO
TFT_VCC
Fajl: SPI_to_UART.kicad_sch
Display Connector
J7
Conn_01x15_Pin
TFT_RST
TFT_VCC_IO
TFT_VCC
Sensors_3.3V
R53 33ohm
SW1
RKJXM1015001
COM
PUSH
JOY_PUSH
JOY_UP
JOY_LEFT
JOY_DOWN
JOY_RIGHT
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Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 2/22

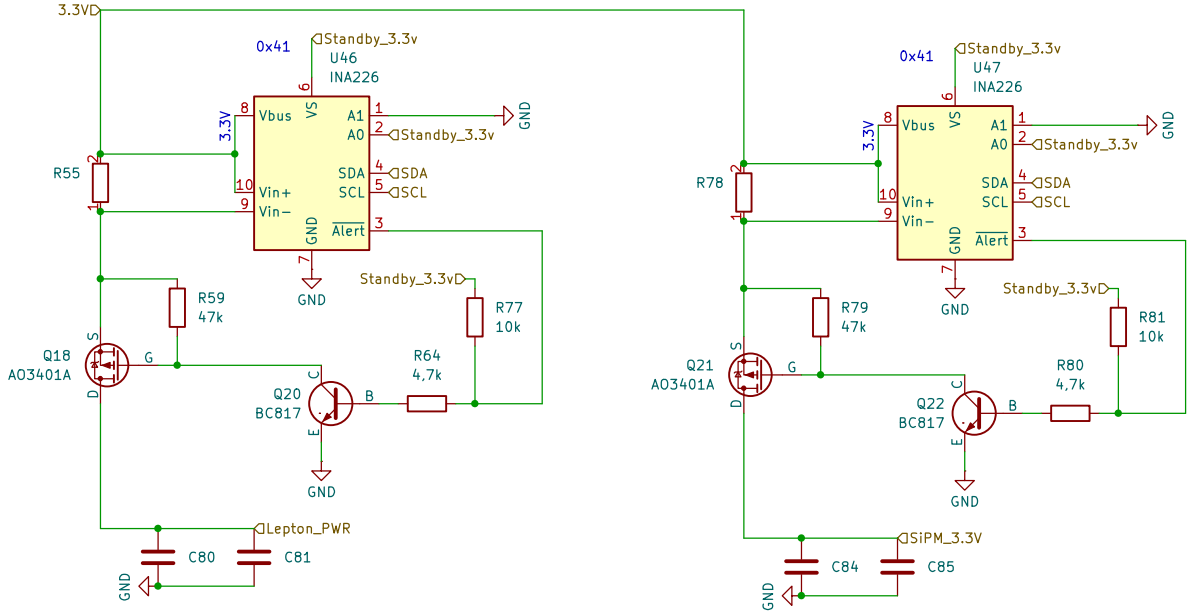


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KiCad E.D.A. 10.0.2-rc1		Id: 3/22

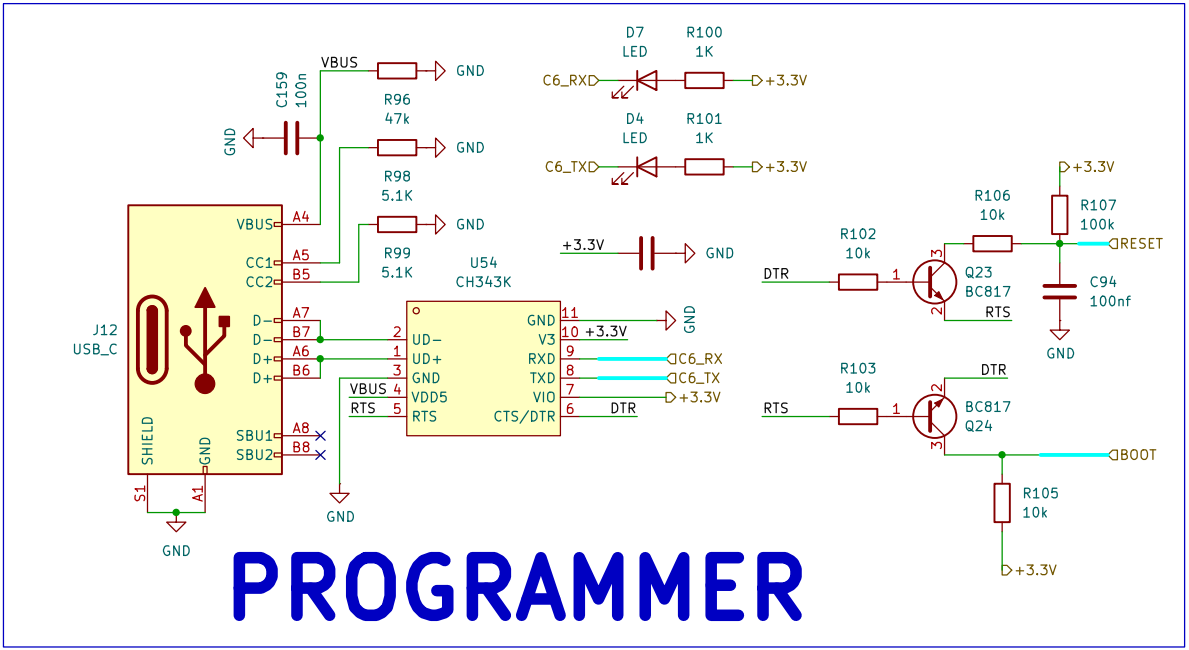


Sheet: /Panel1_MCU/I2C_Sensors/		
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 4/22

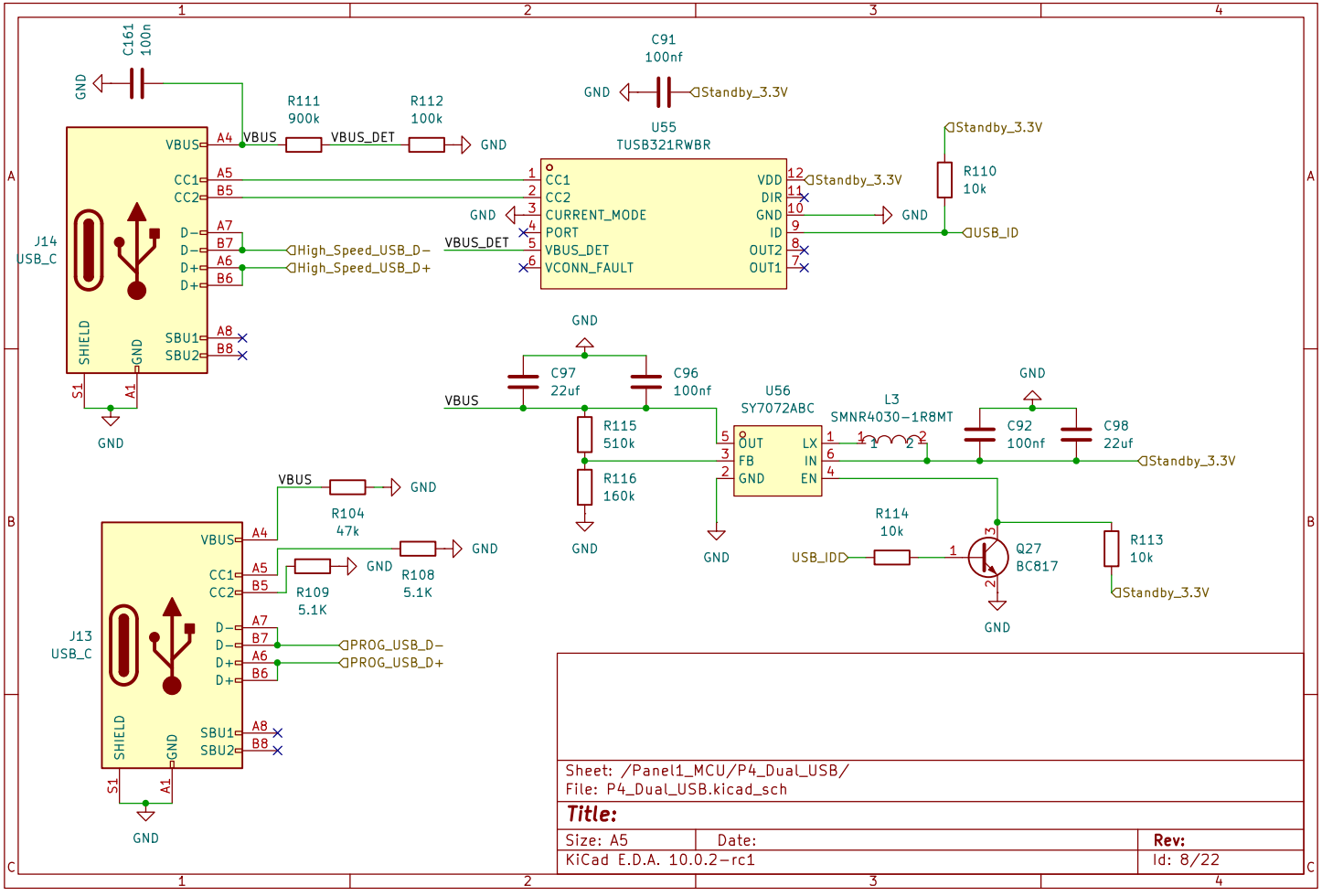

U48
MMC5983MA
SCL/SPI_SCK
VDD
NC
SPI_CS
SPI_SDO
NC
NC
NC
SDA/SPI_SDI
INT
NC
VDDIO
NC
GND
CAP
GND
SCKD
3.3VD
BARO_LP_CSD
MISOD
MOSI
3.3V
C90
22uf
GND
GND
U52
H3LIS331DLTR
Vdd_IO
NC
NC
SCL/SPS
GND
SDA/SDI/SDO
SDO/SA0
CS
GND
Reserved
Vdd
GND
GND
INT1
Reserved
INT2
3.3VD
SCKD
MOSID
MISOD
G-FORCE_LP_CSD
GND
3.3V
GND
GND
GND
U8
MICS-6814
heater_voltage
R88
20ohm
R90
12.4 ohm
R89
42.2ohm
NH3
CO
NO2
D
C
B
A
K
NH3_out
CH_out
heater_voltage
NO2_out
GND
R91
47k
R92
47k
R93
47k
NH3_out
CH_out
NO2_out
GND
U15
TPS7A2028DBV
IN
EN
GND
OUT
3.3VD
C127
2.2uf
C88
2.2uf
C89
100n
heater_voltage
GND
GND
GND
MT1
ICM-42670-P
FSYNC
AP_CS
AP_SCL/AP_SCLK
AP_SDA/AP_SDIO/AP_SDI
AP_SDO/AP_ADO
VDD
VDDIO
INT1/INT
INT2
RESV
GND
ICM_FSYNCD
ICM_LP_CSD
SCKD
MOSID
MISOD
3.3V
GND
U1
AD7091R-8BCPZ-RL7
RESET
CONVST
ADCIN
VIN0
VIN1
VIN2
VIN3
VIN4
VIN5
VIN6
VIN7
VDD
VDRIVE
REGCAP
REFIN/REFOUT
MUXOUT
ALERT/BUSY/GPO0
GPO1
SDO
SDI
SCLK
CS
EPAD
GND
3.3VD
MICS_CONVSTD
ADCIN
NH3_out
CH_out
NO2_out
3.3V
100n
C158
GND
C87
2.2uf
GND
heater_voltage
ADCIN
MISO
MOSI
SCK
MICS_LP_CS
GND
Sheet: /Panel1_MCU/SPI_Sensors/
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Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 5/22

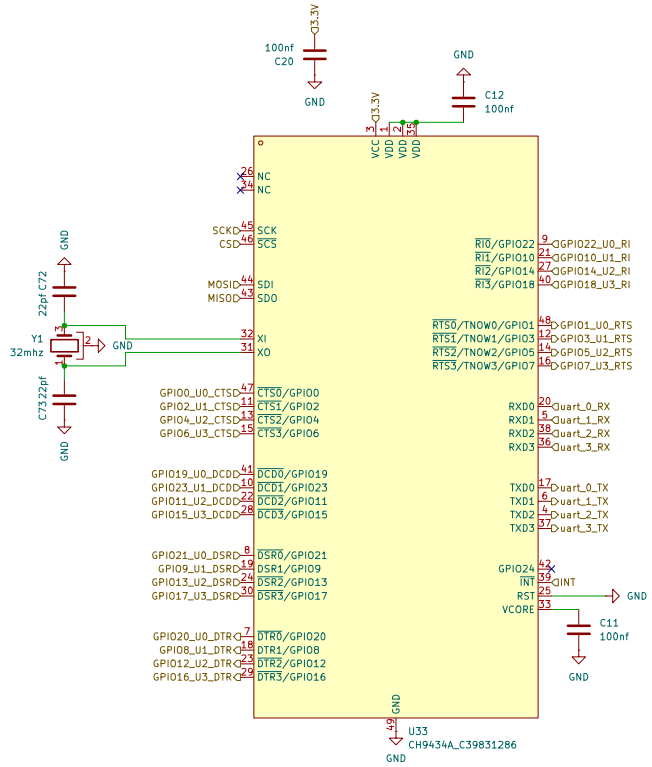


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Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 6/22

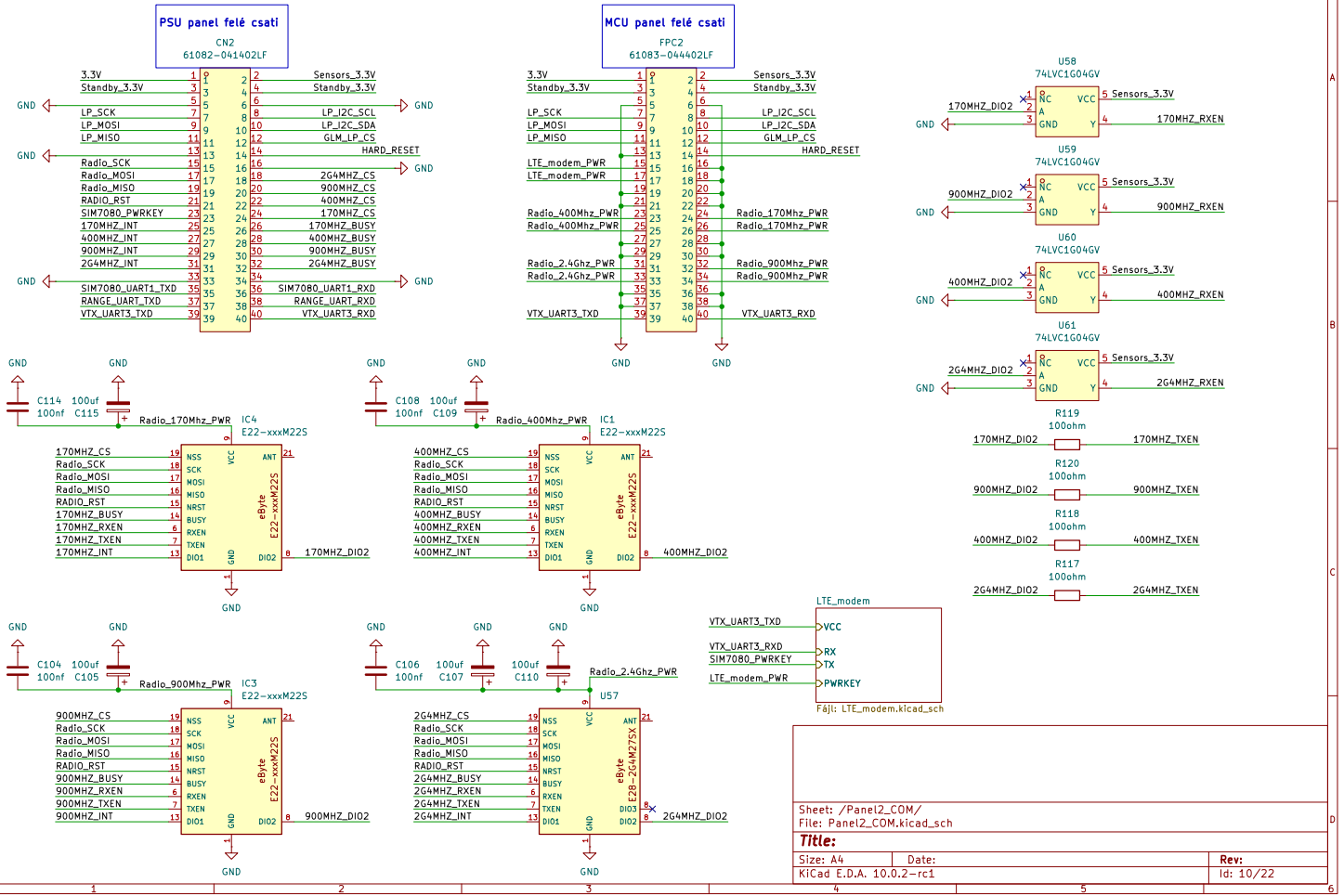


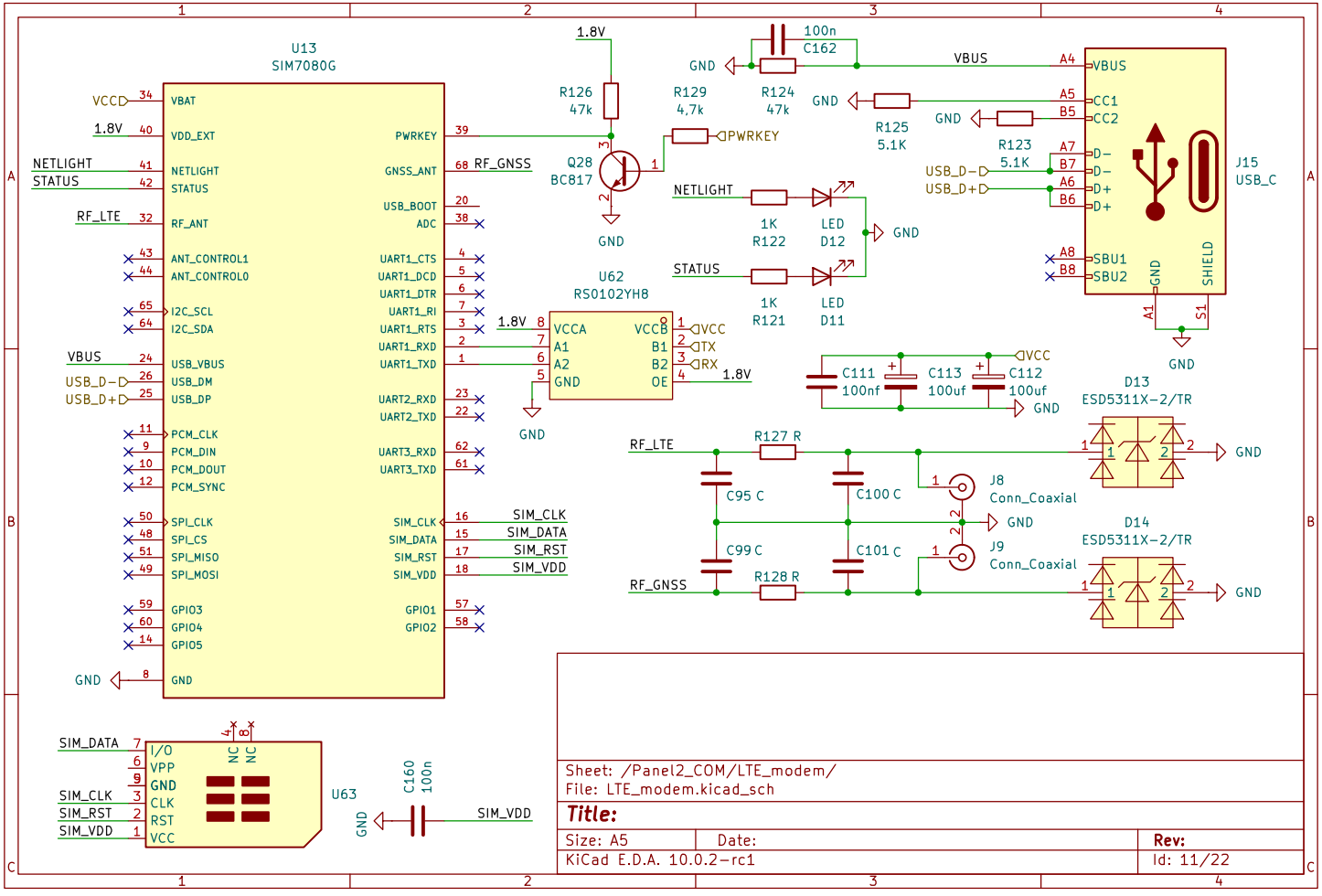
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Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.2−rc1		Id: 7/22

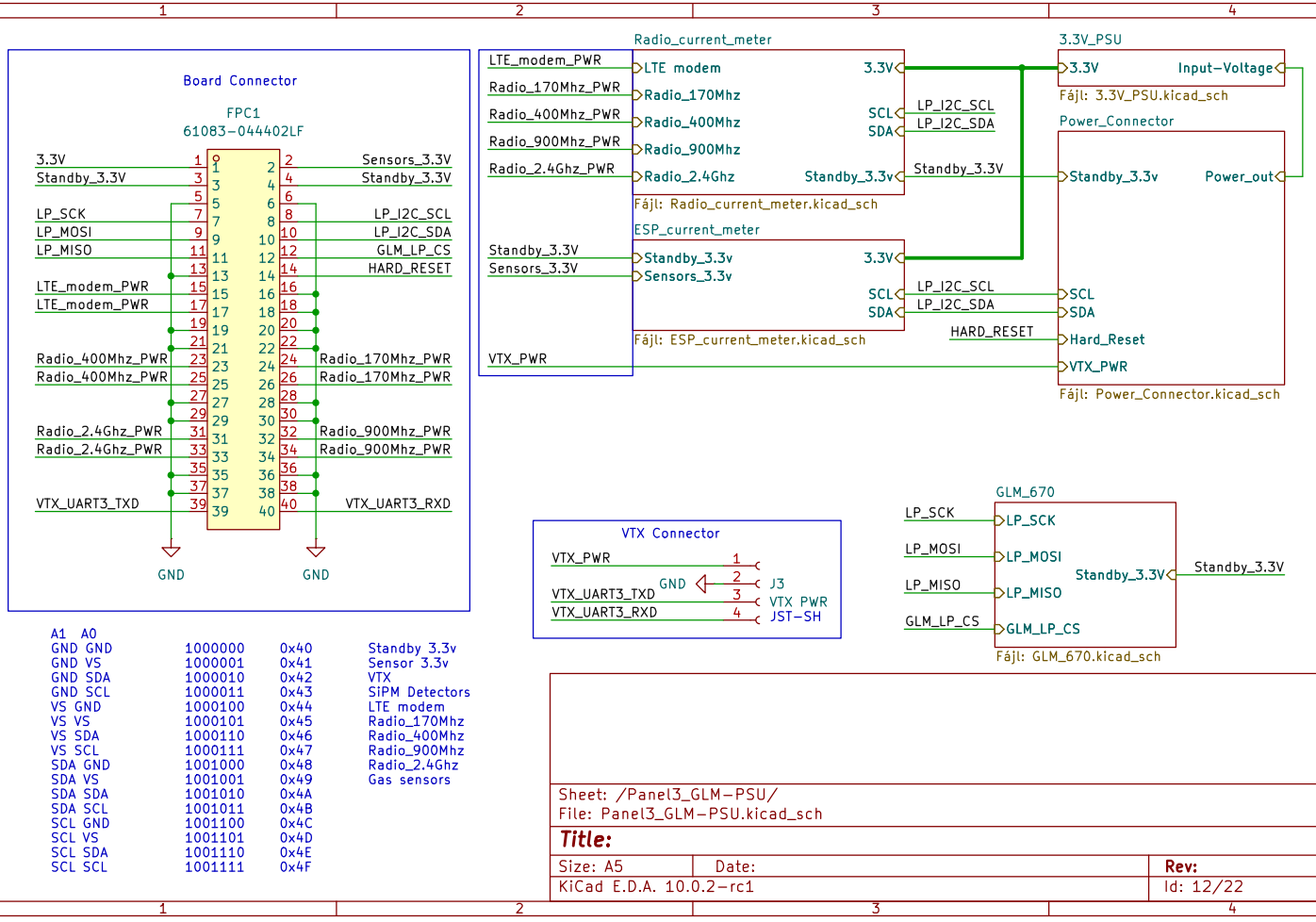

C161
100n
GND
R111
900k
R112
100k
VBUS
VBUS_DET
GND
C91
100nf
GND
Standby_3.3V
U55
TUSB321RWBR
J14
USB_C
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
S1
A1
A4
A5
B5
A7
B7
A6
B6
A8
B8
High_Speed_USB_D-
High_Speed_USB_D+
CC1
CC2
CURRENT_MODE
PORT
VBUS_DET
VCONN_FAULT
VDD
DIR
GND
ID
OUT2
OUT1
R110
10k
USB_ID
C97
22uf
C96
100nf
R115
510k
R116
160k
U56
SY7072ABC
OUT
FB
GND
LX
IN
EN
L3
SMNR4030-1R8MT
C92
100nf
C98
22uf
R104
47k
R109
5.1K
R108
5.1K
J13
USB_C
PROG_USB_D-
PROG_USB_D+
R114
10k
Q27
BC817
R113
10k
USB_ID
Sheet: /Panel1_MCU/P4_Dual_USB/
File: P4_Dual_USB.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 8/22



Sheet: /Panel1_MCU/SPI_to_UART/		
File: SPI_to_UART.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 9/22


PSU panel felé csati
CN2
61082-041402LF
3.3V
Standby_3.3V
GND
LP_SCK
LP_MOSI
LP_MISO
GND
Radio_SCK
Radio_MOSI
Radio_MISO
RADIO_RST
SIM7080_PWRKEY
170MHZ_INT
400MHZ_INT
900MHZ_INT
2G4MHZ_INT
GND
SIM7080_UART1_TXD
RANGE_UART_TXD
VTX_UART3_TXD
Sensors_3.3V
Standby_3.3V
GND
LP_I2C_SCL
LP_I2C_SDA
GLM_LP_CS
HARD_RESET
GND
2G4MHZ_CS
900MHZ_CS
400MHZ_CS
170MHZ_CS
170MHZ_BUSY
400MHZ_BUSY
900MHZ_BUSY
2G4MHZ_BUSY
GND
SIM7080_UART1_RXD
RANGE_UART_RXD
VTX_UART3_RXD
MCU panel felé csati
FPC2
61083-044402LF
3.3V
Standby_3.3V
LP_SCK
LP_MOSI
LP_MISO
LTE_modem_PWR
LTE_modem_PWR
Radio_400Mhz_PWR
Radio_400Mhz_PWR
Radio_2.4Ghz_PWR
Radio_2.4Ghz_PWR
VTX_UART3_TXD
GND
Sensors_3.3V
Standby_3.3V
LP_I2C_SCL
LP_I2C_SDA
GLM_LP_CS
HARD_RESET
Radio_170Mhz_PWR
Radio_170Mhz_PWR
Radio_900Mhz_PWR
Radio_900Mhz_PWR
VTX_UART3_RXD
GND
U58
74LVC1G04GV
170MHZ_DIO2
GND
Sensors_3.3V
170MHZ_RXEN
U59
74LVC1G04GV
900MHZ_DIO2
GND
Sensors_3.3V
900MHZ_RXEN
U60
74LVC1G04GV
400MHZ_DIO2
GND
Sensors_3.3V
400MHZ_RXEN
U61
74LVC1G04GV
2G4MHZ_DIO2
GND
Sensors_3.3V
2G4MHZ_RXEN
R119
100ohm
170MHZ_DIO2
170MHZ_TXEN
R120
100ohm
900MHZ_DIO2
900MHZ_TXEN
R118
100ohm
400MHZ_DIO2
400MHZ_TXEN
R117
100ohm
2G4MHZ_DIO2
2G4MHZ_TXEN
GND
C114 100uf
100nf C115
Radio_170Mhz_PWR
IC4
E22-xxxM22S
170MHZ_CS
Radio_SCK
Radio_MOSI
Radio_MISO
RADIO_RST
170MHZ_BUSY
170MHZ_RXEN
170MHZ_TXEN
170MHZ_INT
NSS
SCK
MOSI
MISO
NRST
BUSY
RXEN
TXEN
DIO1
VCC
ANT
GND
DIO2
eByte
E22-xxxM22S
170MHZ_DIO2
GND
C108 100uf
100nf C109
Radio_400Mhz_PWR
IC1
E22-xxxM22S
400MHZ_CS
Radio_SCK
Radio_MOSI
Radio_MISO
RADIO_RST
400MHZ_BUSY
400MHZ_RXEN
400MHZ_TXEN
400MHZ_INT
400MHZ_DIO2
GND
C104 100uf
100nf C105
Radio_900Mhz_PWR
IC3
E22-xxxM22S
900MHZ_CS
Radio_SCK
Radio_MOSI
Radio_MISO
RADIO_RST
900MHZ_BUSY
900MHZ_RXEN
900MHZ_TXEN
900MHZ_INT
900MHZ_DIO2
GND
C106 100uf
100nf C107
100uf
C110
Radio_2.4Ghz_PWR
U57
2G4MHZ_CS
Radio_SCK
Radio_MOSI
Radio_MISO
RADIO_RST
2G4MHZ_BUSY
2G4MHZ_RXEN
2G4MHZ_TXEN
2G4MHZ_INT
DIO3
E28-2G4M27SX
2G4MHZ_DIO2
GND
LTE_modem
VTX_UART3_TXD
VTX_UART3_RXD
SIM7080_PWRKEY
LTE_modem_PWR
VCC
RX
TX
PWRKEY
Fájl: LTE_modem.kicad_sch
Sheet: /Panel2_COM/
File: Panel2_COM.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 10/22


U13
SIM7080G
VCC
1.8V
NETLIGHT
STATUS
RF_LTE
VBUS
USB_D-
USB_D+
GND
VBAT
VDD_EXT
NETLIGHT
STATUS
RF_ANT
ANT_CONTROL1
ANT_CONTROL0
I2C_SCL
I2C_SDA
USB_VBUS
USB_DM
USB_DP
PCM_CLK
PCM_DIN
PCM_DOUT
PCM_SYNC
SPI_CLK
SPI_CS
SPI_MISO
SPI_MOSI
GPIO3
GPIO4
GPIO5
GND
PWRKEY
GNSS_ANT
USB_BOOT
ADC
UART1_CTS
UART1_DCD
UART1_DTR
UART1_RI
UART1_RTS
UART1_RXD
UART1_TXD
UART2_RXD
UART2_TXD
UART3_RXD
UART3_TXD
SIM_CLK
SIM_DATA
SIM_RST
SIM_VDD
GPIO1
GPIO2
RF_GNSS
R126
47k
Q28
BC817
PWRKEY
R129
4,7k
R124
47k
100n
C162
R125
5.1K
R123
5.1K
J15
USB_C
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
SHIELD
NETLIGHT
STATUS
1K
R122
1K
R121
LED
D12
LED
D11
U62
RS0102YH8
VCCA
A1
A2
GND
VCCB
B1
B2
OE
VCC
TX
RX
1.8V
C111
100nf
C113
100uf
C112
100uf
D13
ESD5311X-2/TR
D14
ESD5311X-2/TR
RF_LTE
R127 R
C95 C
C100 C
C99 C
C101 C
R128 R
RF_GNSS
J8
Conn_Coaxial
J9
Conn_Coaxial
U63
I/O
VPP
GND
CLK
RST
VCC
NC
NC
C160
100n
SIM_DATA
SIM_CLK
SIM_RST
SIM_VDD
Sheet: /Panel2_COM/LTE_modem/
File: LTE_modem.kicad_sch
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Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 11/22

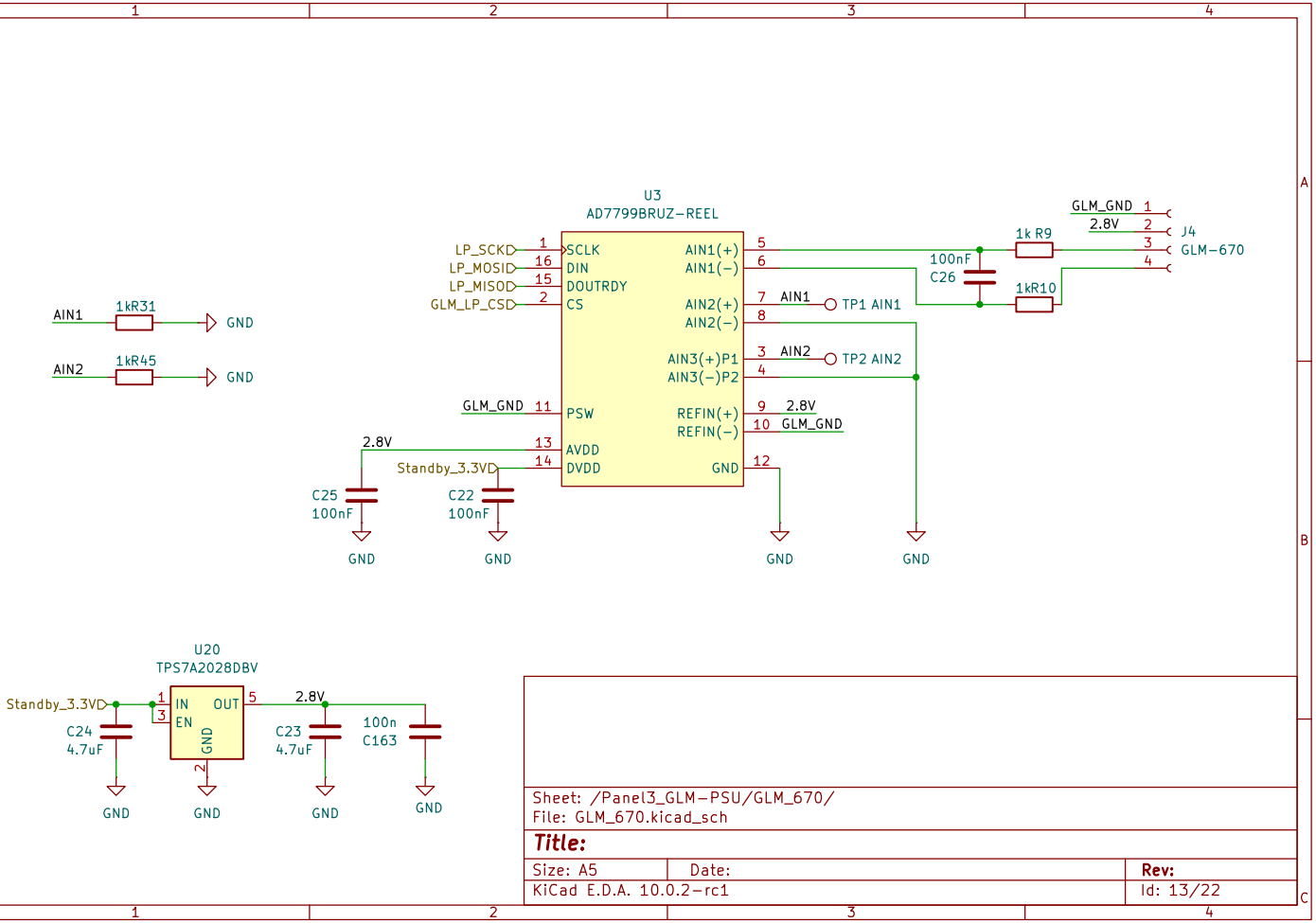


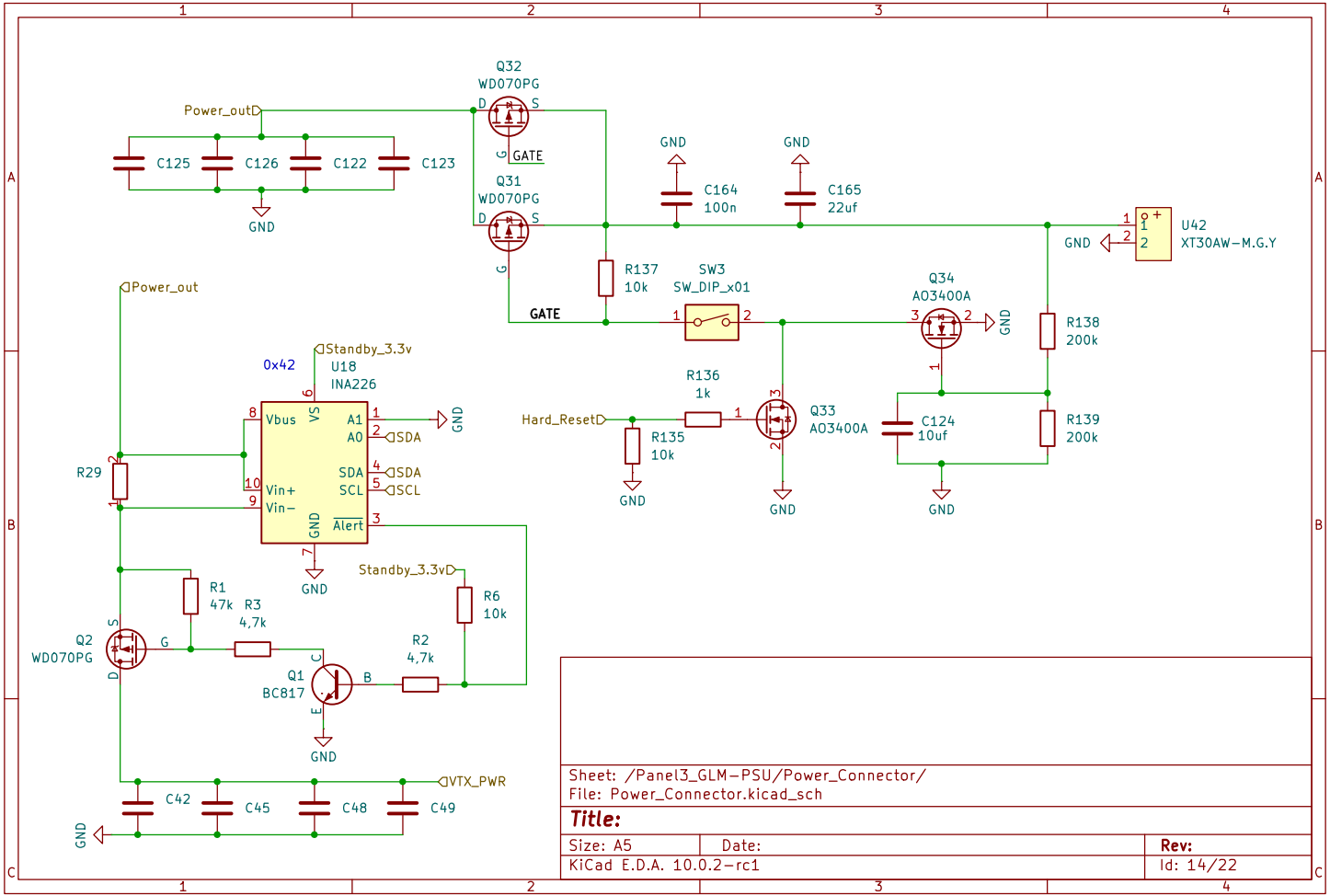
A1	A0			
GND	GND	1000000	0x40	Standby 3.3v
GND	VS	1000001	0x41	Sensor 3.3v
GND	SDA	1000010	0x42	VTX
GND	SCL	1000011	0x43	SiPM Detectors
VS	GND	1000100	0x44	LTE modem
VS	VS	1000101	0x45	Radio_170Mhz
VS	SDA	1000110	0x46	Radio_400Mhz
VS	SCL	1000111	0x47	Radio_900Mhz
SDA	GND	1001000	0x48	Radio_2.4Ghz
SDA	VS	1001001	0x49	Gas sensors
SDA	SDA	1001010	0x4A	
SDA	SCL	1001011	0x4B	
SCL	GND	1001100	0x4C	
SCL	VS	1001101	0x4D	
SCL	SDA	1001110	0x4E	
SCL	SCL	1001111	0x4F	

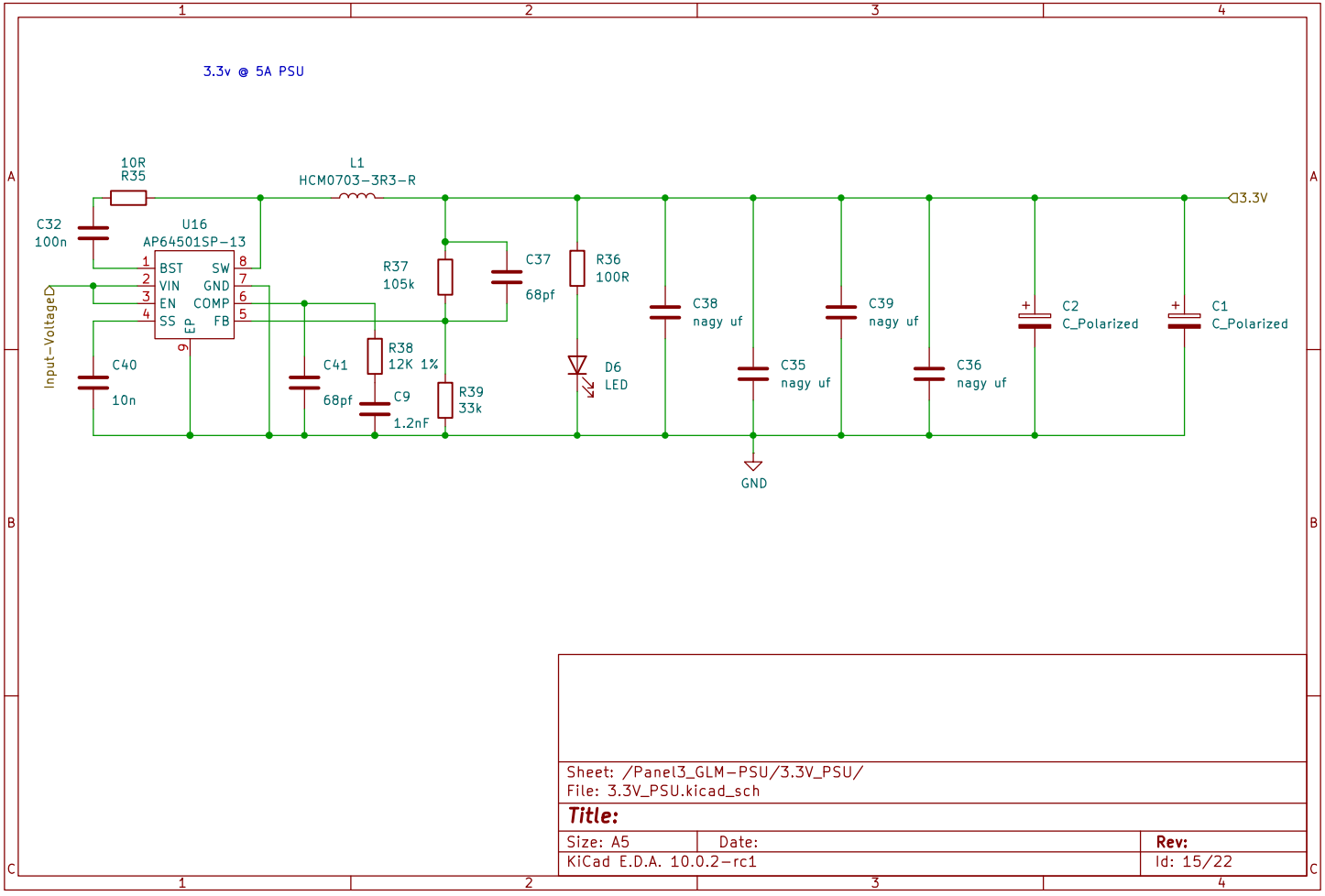
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File: Panel3_GLM-PSU.kicad_sch

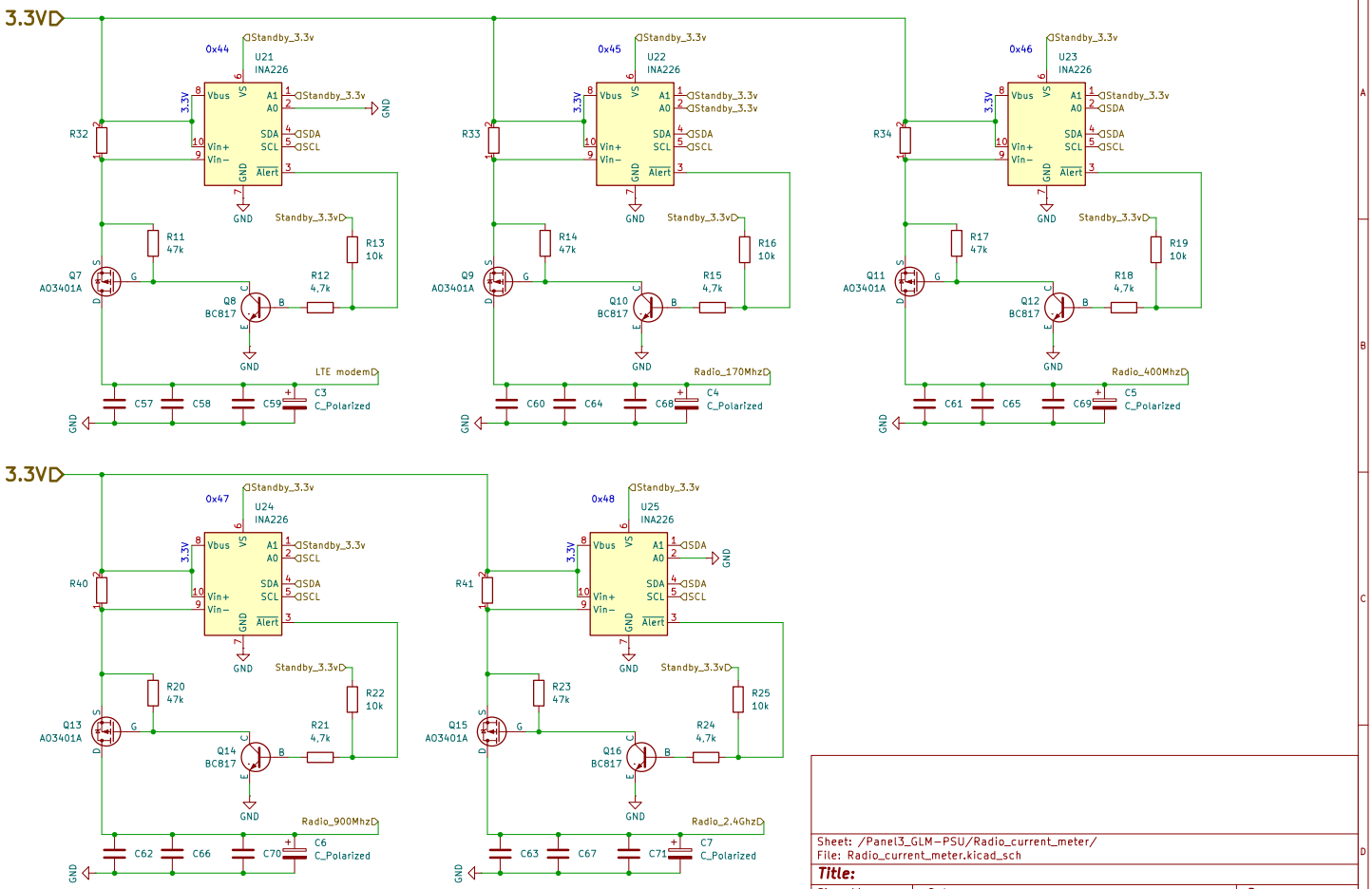
Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 12/22


U3
AD7799BRUZ-REEL
LP_SCK
LP_MOSI
LP_MISO
GLM_LP_CS
SCLK
DIN
DOUTRDY
CS
PSW
AVDD
DVDD
AIN1(+)
AIN1(-)
AIN2(+)
AIN2(-)
AIN3(+)P1
AIN3(-)P2
REFIN(+)
REFIN(-)
GND
AIN1
AIN2
TP1 AIN1
TP2 AIN2
2.8V
GLM_GND
Standby_3.3V
AIN1
1kR31
GND
AIN2
1kR45
GND
1k R9
1kR10
100nF
C26
GLM_GND
2.8V
J4
GLM-670
C25
100nF
C22
100nF
U20
TPS7A2028DBV
Standby_3.3V
IN
OUT
EN
GND
C24
4.7uF
C23
4.7uF
100n
C163
Sheet: /Panel3_GLM-PSU/GLM_670/
File: GLM_670.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 13/22


Power_out
C125
C126
C122
C123
GND
Q32
WD070PG
Q31
WD070PG
GATE
C164
100n
C165
22uf
U42
XT30AW-M.G.Y
R137
10k
SW3
SW_DIP_x01
GATE
Q34
AO3400A
R138
200k
R139
200k
C124
10uf
R136
1k
Hard_Reset
R135
10k
Q33
AO3400A
Standby_3.3v
0x42
U18
INA226
Vbus
Vin+
Vin−
VS
GND
A1
A0
SDA
SCL
Alert
R29
R1
47k
R3
4,7k
R2
4,7k
R6
10k
Q2
WD070PG
Q1
BC817
VTX_PWR
C42
C45
C48
C49
Sheet: /Panel3_GLM-PSU/Power_Connector/
File: Power_Connector.kicad_sch
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Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 14/22

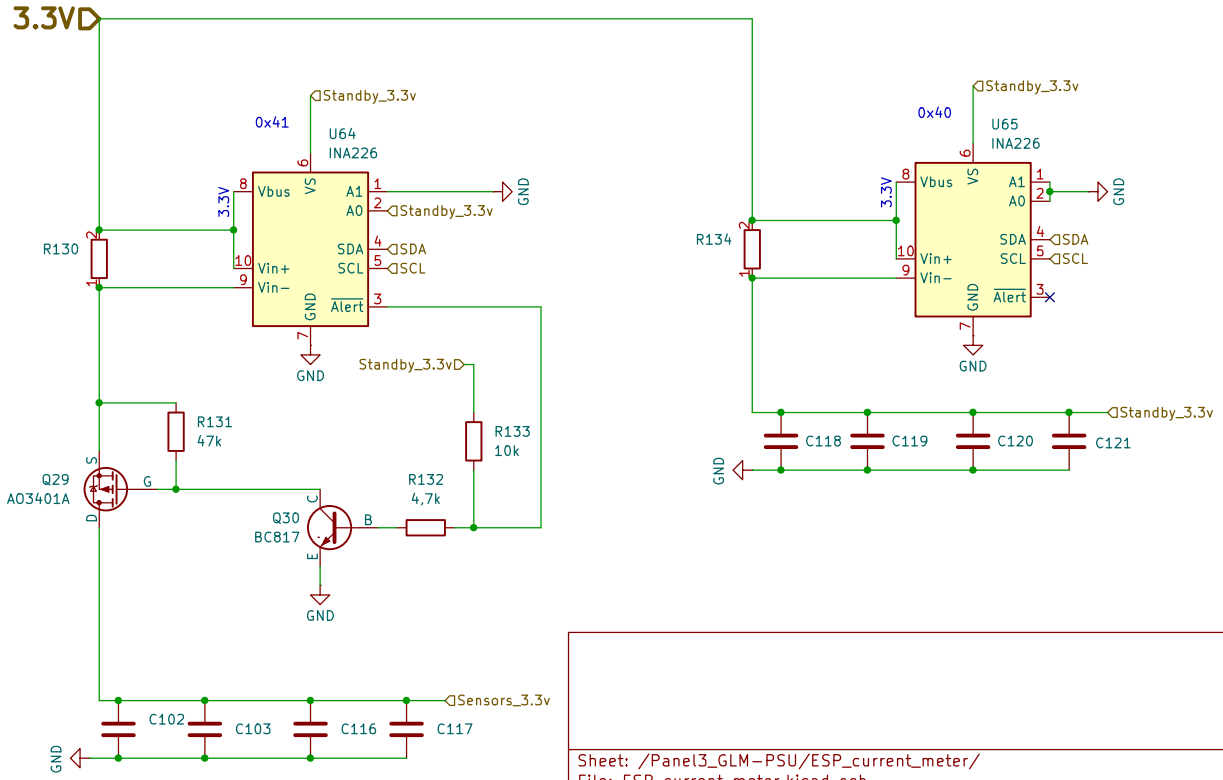

3.3v @ 5A PSU
10R
R35
L1
HCM0703-3R3-R
C32
100n
U16
AP64501SP-13
BST
VIN
EN
SS
SW
GND
COMP
FB
EP
9
Input-Voltage
C40
10n
R37
105k
C37
68pf
R36
100R
C38
nagy uf
C39
nagy uf
C2
C_Polarized
C1
C_Polarized
3.3V
C41
68pf
R38
12K 1%
C9
1.2nF
R39
33k
D6
LED
C35
nagy uf
C36
nagy uf
GND
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File: 3.3V_PSU.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 15/22



Sheet: /Panel3_GLM-PSU/Radio_current_meter/
File: Radio_current_meter.kicad_sch

Title:

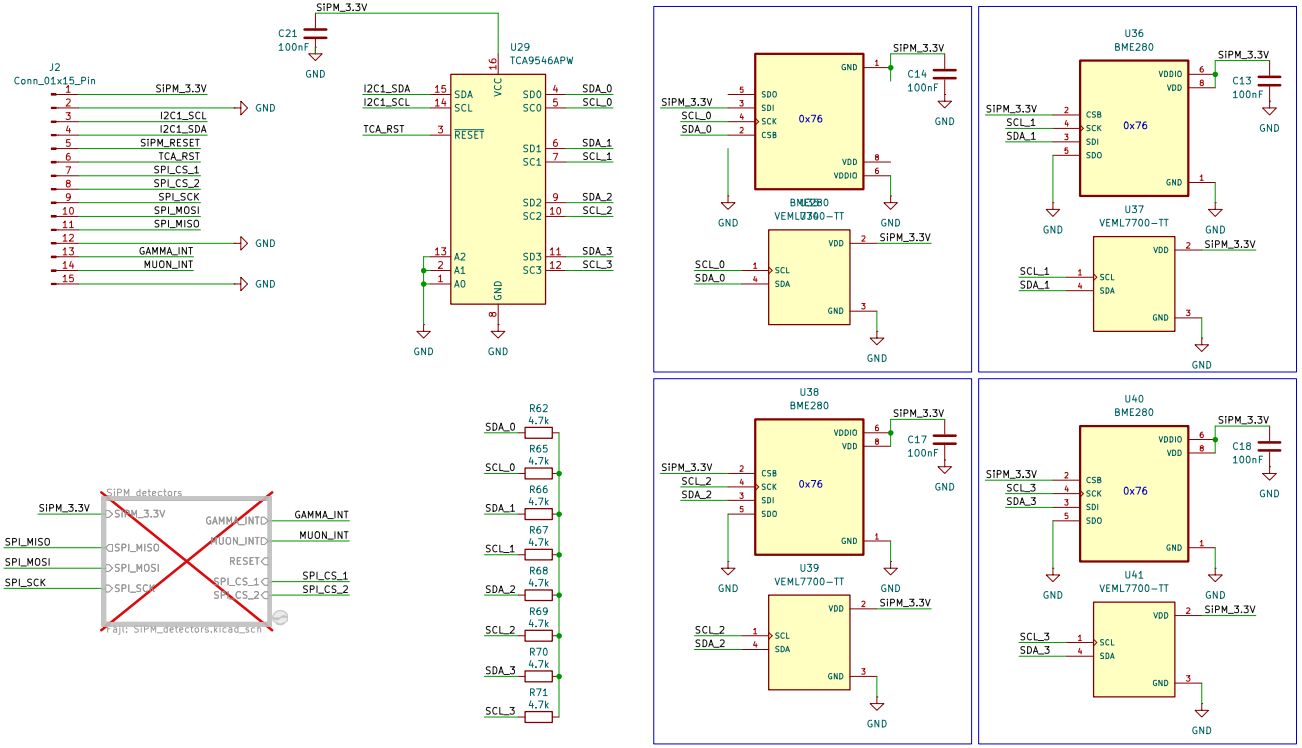
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 16/22



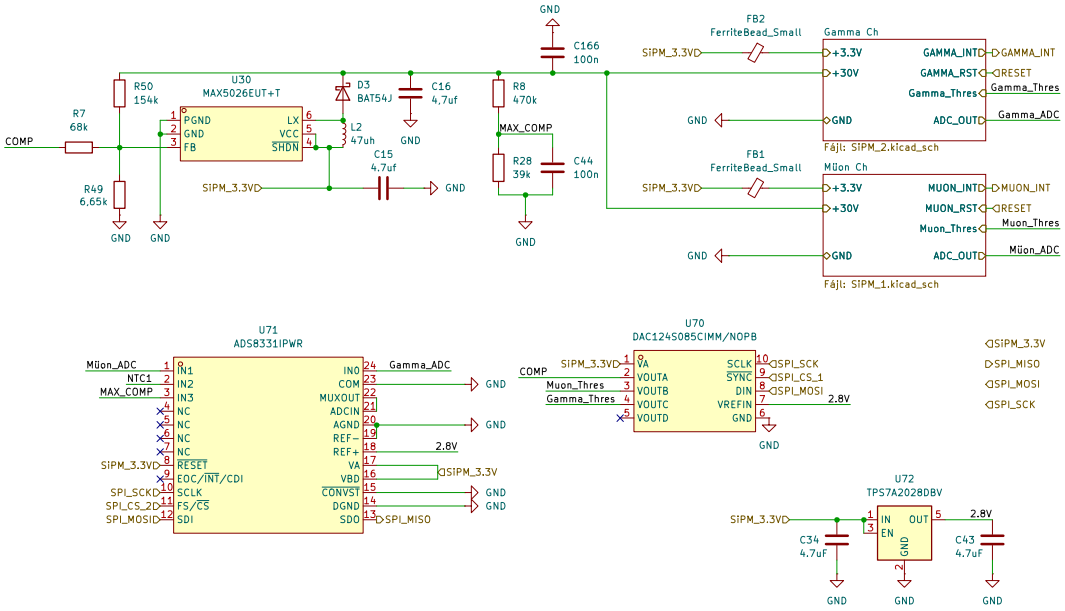
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File: ESP_current_meter.kicad_sch

Title:

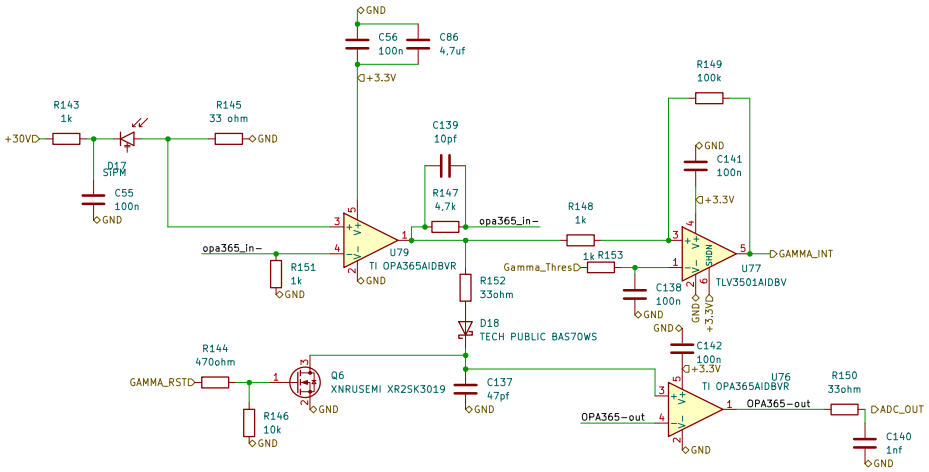
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 17/22



Sheet: /Panel4_BAT_Plate/		
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 18/22



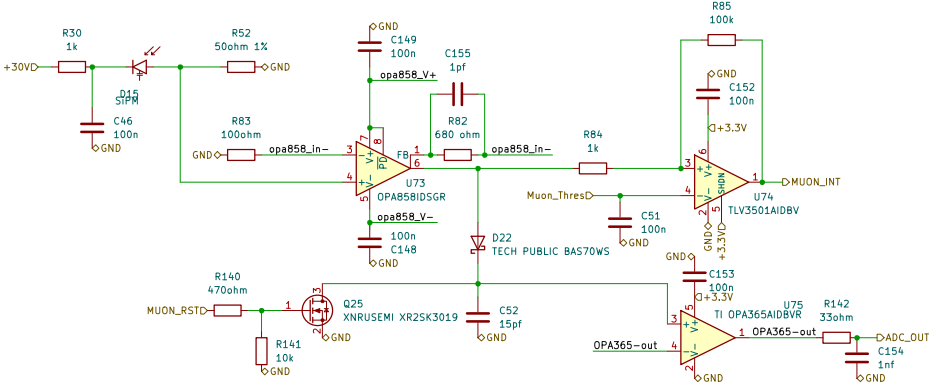
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 19/22



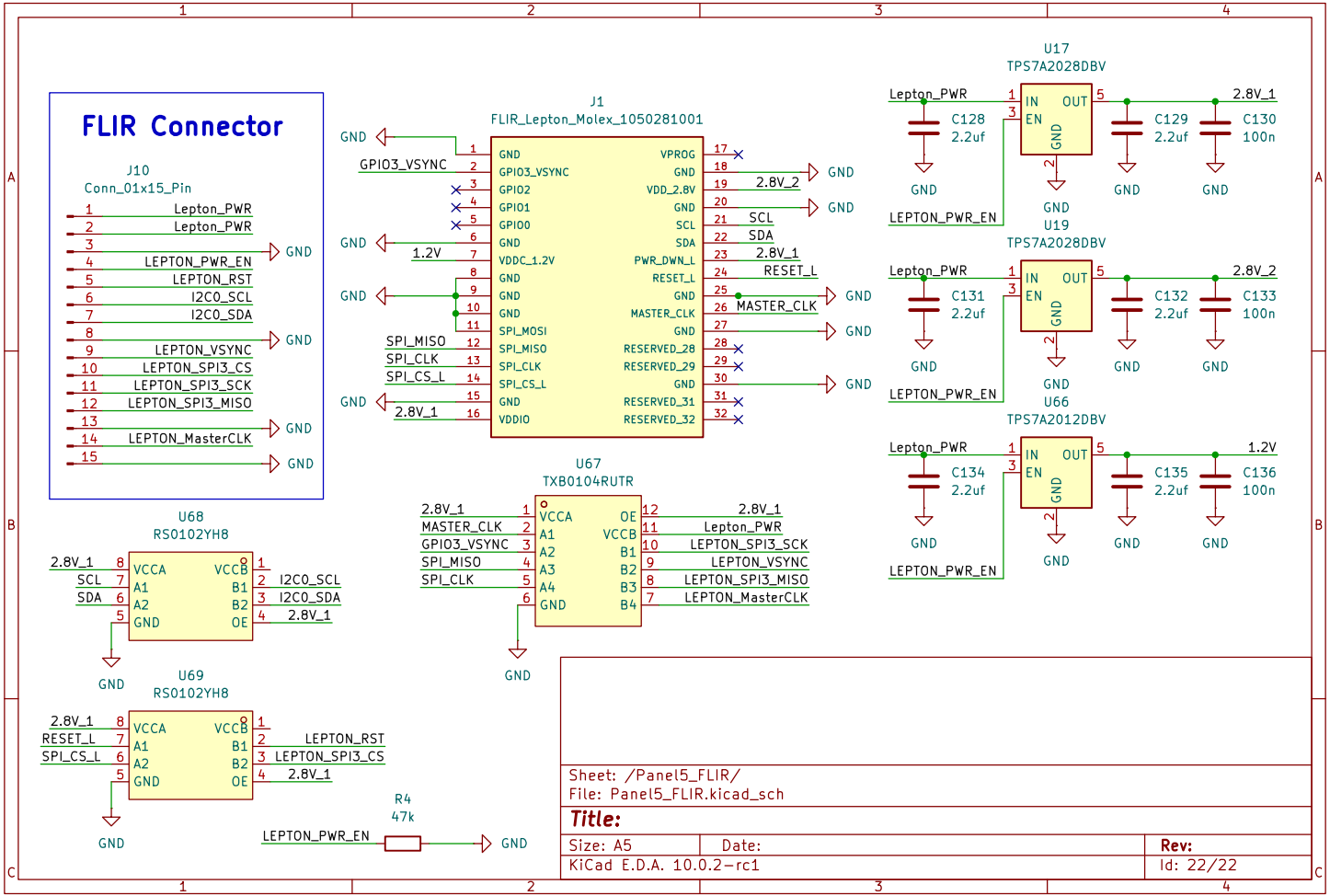
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File: SiPM_2.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 20/22



Sheet: /Panel4_BAT_Plate/SiPM_detectors/Müon Ch/		
File: SiPM_1.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 21/22


FLIR Connector
J10
Conn_01x15_Pin
1 Lepton_PWR
2 Lepton_PWR
3 GND
4 LEPTON_PWR_EN
5 LEPTON_RST
6 I2C0_SCL
7 I2C0_SDA
8 GND
9 LEPTON_VSYNC
10 LEPTON_SPI3_CS
11 LEPTON_SPI3_SCK
12 LEPTON_SPI3_MISO
13 GND
14 LEPTON_MasterCLK
15 GND
J1
FLIR_Lepton_Molex_1050281001
GND
GPIO3_VSYNC
1.2V
SPI_MISO
SPI_CLK
SPI_CS_L
2.8V_1
2.8V_2
SCL
SDA
RESET_L
MASTER_CLK
RESERVED_28
RESERVED_29
RESERVED_31
RESERVED_32
U17
TPS7A2028DBV
U19
TPS7A2028DBV
U66
TPS7A2012DBV
Lepton_PWR
LEPTON_PWR_EN
C128 2.2uf
C129 2.2uf
C130 100n
C131 2.2uf
C132 2.2uf
C133 100n
C134 2.2uf
C135 2.2uf
C136 100n
2.8V_1
2.8V_2
1.2V
U67
TXB0104RUTR
U68
RS0102YH8
U69
RS0102YH8
R4
47k
Sheet: /Panel5_FLIR/
File: Panel5_FLIR.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 22/22